

Reviewer Pack — Tropical Geometry of Boolean Functions and AC0 Parity Circui...

Entry 0fd5ae62f864 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-24 01:24:33 UTC

Tropical Geometry of Boolean Functions and AC0 Parity Circuit Size

Entry ID: 0fd5ae62f864 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-24 01:24:33 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry **Field B** (complexity object): AC0 Parity Complexity

Statement:

[‘For any Boolean function f with n inputs, the size of its smallest AC0 parity circuit C_f is upper-bounded by $O(\text{TropDiameter}(f) * \log(n))$, where $\text{TropDiameter}(f)$ is the tropical diameter of f .’, ‘The tropical diameter of a Boolean function f is defined as the maximum distance between any two points in the real projective plane that correspond to different inputs of f under the tropical metric.’]

Rationale (proposer’s reasoning):

[“Tropical geometry provides a way to study the structure of Boolean functions and their representations. By leveraging the tropical diameter, which measures the complexity of the function’s input space, we can potentially derive a lower bound on the size of parity circuits.”, ‘This conjecture builds upon previous work that has shown connections between tropical geometry and circuit complexity, suggesting that tropical properties could be used to characterize AC0 complexity.’]

Taxonomy category: AC0_PARITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 4459c51d7c5eb3c0

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a Boolean function f with n inputs, C_f is considered supported if $C_f \leq O(\text{TropDiameter}(f) * \log(n))$ for all seeds and falsified if any seed produces $C_f > O(\text{TropDiameter}(f) * \log(n))$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "tropical geometry" AND "AC0 parity complexity" - "Boolean function" AND "tropical diameter" AND AC0 - "smallest AC0 circuit size" AND "upper bound" AND tropical geometry

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.3s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.randint(0, 1) for _ in range(2**n)]

    def tropical_diameter(f):
        n = int(math.log2(len(f)))
        points = [(i, f[i]) for i in range(len(f))]
        max_distance = 0
        for i in range(len(points)):
            for j in range(i + 1, len(points)):
                distance = sum(abs(a - b) for a, b in zip(points[i], points[j]))
                if distance > max_distance:
                    max_distance = distance
        return max_distance

    def ac0_circuit_size(f):
        n = int(math.log2(len(f)))
        # Simplified AC0 circuit size estimation (not actual implementation)
        return 2**n

    f = generate_boolean_function(random.randint(5, 40))
    trop_diameter = tropical_diameter(f)
    ac0_size = ac0_circuit_size(f)

    metric_value = ac0_size
```

```

instances_tested = 1
conjecture_holds = (ac0_size <= trop_diameter * math.log(len(f)))
counterexample = "" if conjecture_holds else f"Function {f} with n={len(f)}, TropDiameter={tro

return {
    "metric_name": "AC0 Circuit Size",
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    if len(sys.argv[1:]) > 0:
        seeds = [int(s) for s in sys.argv[1:]]
    else:
        # Default list of 30 primes
        seeds = [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 7

results = []
for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_value = sum(r["metric_value"] for r in results) / len(results)
std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fra
elif any(not r["conjecture_holds"] for r in results):
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing
else:
    print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means we cannot confirm whether the conjecture holds for all seeds. | next: Run the test with a longer timeout or increase the number of seeds to ensure that the conjecture is tested thoroughly.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	9696
2	preregistration	ollama_remote	glm4:latest	0	0	5665
3	novelty	ollama_remote	glm4:latest	0	0	4624
4	novelty	ollama_remote	glm4:latest	0	0	5388
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14233
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8641
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7265
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9419
9	judge	ollama_remote	glm4:latest	0	0	8945

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 73876 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/0fd5ae62f864.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/0fd5ae62f864.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/0fd5ae62f864.tar.gz` (if generated)